

EXPERIENCE

AI Engineer, Tesco

Winter 2024 – Present

Tesco Meal Planner Agent

- Delivered production LLM agents that translate open-ended meal requests, dietary needs, preferences, and shopping context into personalised recipes, product selections, and ready-to-shop baskets; shipped **from prototype to beta in 10 weeks** with OpenAI-acquired Tomoro engineers, reaching **380k+ internal users**
- Shipped an **LLM-agent evaluation harness** using Langfuse traces, measuring recipe quality, tool-call correctness, product-selection behaviour, and state transitions with deterministic, LLM-as-a-Judge, and self-consistency evaluators
- Built **layered input and output guardrails** and stress-tested them with retail-specific Promptfoo red-team suites; turned failures into retry policies, prompt updates, and regression-gated improvements

Substitute Product Recommendation

- Built **PySpark data pipelines** over millions of retail records from **10+ sources**, including online clickstream and offline store data, powering product-substitution models across **3k+ stores** and delivering **£5M+ annualised savings**
- Improved cold-start reliability by combining sparse behavioural data with LLM-based semantic similarity over product metadata, increasing substitute coverage by **32.51%**

Off-Cycle Intern, Global Banking & Markets Division, Goldman Sachs

Autumn 2024

- Built Slang pipelines converting unstructured financial streams into structured daily market insights

RESEARCH PROJECTS

SETA: Scaling Environments for Terminal Agents, Second Author, CAMEL-AI

Winter 2025 – Present

- Built an **end-to-end agent training and evaluation pipeline** spanning source-data selection, environment synthesis, task verification, difficulty evolution, reward design, robust toolkit design for terminal agents, GRPO training, and benchmark evaluation
- Built **SETA-Env**, the **largest open-source verifiable terminal-agent environment dataset** to date: **4.5k+** Docker-isolated, executable, and verifiable environments for RLVR training of terminal agents, with 13k+ downloads on Hugging Face
- Built **SETA-Synth** and **SETA-Evol** pipelines to generate diverse task variants, evolve difficulty against model capability for **curriculum learning**, and filter out tasks that were too easy or too hard, preserving useful training signal
- Validated the environments through 8B-scale agentic RL post-training with AReaL and SGLang, improving Qwen3-8B on Terminal-Bench 2.0 from 3.6% to **12.0%**, achieving **SOTA at 8B scale** and **surpassing GPT-5-Nano**, with transfer gains on CompileBench (6.7% → 40%) and CRUST-Bench (15% → 24%)

Blockchain Coder: Multi-Agent Code Generation Framework, UCL

Summer 2024

- Designed a **multi-agent framework** (Planner, Optimizer, Debugger) for code-generation research, improving generalisation beyond memorisation and mitigating the ~40% accuracy drop on unseen coding tasks
- Ran controlled experiments on problem decomposition and unit-test-driven interactive refinement, doubling pass@1 to 6.90% and improving private-test pass rate by **5.7x**

EDUCATION

University College London | MSc Machine Learning | First Class

2023 – 2024

- Multi-agent AI (85.5%), Information Retrieval and Data Mining (87.5%), Bayesian Deep Learning (83.5%)

Imperial College London | BSc Mathematics | First Class (80.3%)

2020 – 2023

- Methods for Data Science (94.4%), Mathematical Foundations of Machine Learning (90%)

TECHNICAL SKILLS

Languages	Python, SQL, R, Julia; Mandarin (Native), Cantonese (Native), English (Fluent)
Applied AI & Evaluation	Docker, New Relic, LLM-as-a-Judge, Langfuse, Promptfoo, LangChain, AReaL
Data Science & ML	Statistical modeling, PySpark, Airflow, ETL pipelines, A/B testing, PyTorch, HuggingFace

ACHIEVEMENTS

- 5x hackathon winner with £8k+ in prizes; projects include Sign2Speak, SquareDeal, and CartPilot